

ECE 1901
Technical Answers For Real World Problems(TARP)
Report

A project report titled
FISHERMAN BORDER ALERT SYSTEM
AND FISH DETECTION

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Submitted to

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School of Electronics Engineering

DECLARATION BY THE CANDIDATE

I hereby declare that the Report entitled “**FISHERMAN BORDER ALERT SYSTEM AND FISH DETECTION**” submitted by me to VIT Chennai is a record of bonafide work undertaken by me under the supervision of **Dr. R. Menaka, Professor, SENSE, VIT Chennai.**

Signature of the Candidate

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21/04/2022.

ACKNOWLEDGEMENT

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We thank our parents, family, and friends for bearing with us throughout the course of our project and for the opportunity they provided us in undergoing this course in such a prestigious institution.

BONAFIDE CERTIFICATE

Certified that this project report entitled “FISHERMAN BORDER ALERT SYSTEM AND FISH DETECTION” is a bonafide work of **MANSI BHARADWAJ (19BEC1051)** , **NANDESH P G (19BEC1128)** , **SHANMUKH SAI (19BEC1062)** , **NARENDRA KUMAR (19BEC1270)** and **KRISHNAA YOGITH R S (19BEC1262)** carried out the “J”-Project work under my supervision and guidance for CSE2003 Data Structures and Algorithms.

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ABSTRACT

There have been several cases where the ignorant and poor fishermen of Tamil Nadu have unknowingly and due to lack of technical equipment and knowledge have crossed the international border of sea and has been arrested by Sri Lankan naval forces on the accusation of terrorism. These innocent fishermen with no fault of theirs get caught and get severe punishment for the fault they don't know about. Countries with the International Maritime Boundary Line (IMBL) will always have security problems and continuous life threats for those fishermen whose family's main economic support is fishing. Even in the peninsular countries like India have their boundary limit in the ocean, the people of these coastal regions have the main work of fishing, due to carelessness or without knowing their boundary limit of their country they cross the borders. Nowadays catching fish has become difficult due to the steady decline in the fish population due to pollution and overfishing. Fishermen spend a lot of money and time to go fishing in deep waters. Many of the time they won't catch many fishes due to lack of knowledge about the fish movements. This is also one of the reasons fishermen cross borders in hope of catching more fishes.

So, in order to solve these two problems, our group took this as a challenge and We have found an easy and feasible way to solve this problem in which the fishermen would be given certain signals and a buzzer would sound as soon as they approach the border and thus, they can remain at a safe distance from the border. We have included an underwater ultrasonic sensor which gives the distance data of the fish which we process using processing software to visually show the fishermen the density of the fishes. This would greatly help the poor and innocent fishermen to remain within the Indian borders and eases the fishing process.

CHAPTER – 1

INTRODUCTION

The Tamil Nadu fishermen even today invoke the historical rights and routinely stay into the International Maritime Boundary Line (IMBL) for fishing. From Tamil Nadu about 18,000 boats of different kinds conduct fishing along the India-Sri Lanka maritime border. But by accidentally crossing the border without knowledge, they get shot by the Lankan navy. This leads to loss in both humans as well as their economic incomes. We have developed a system which eliminates such problems and saves the lives of the fishermen. GPS commonly known as Global Positioning System also known as Navistar GPS that provides geolocation and the information to a GPS receiver in all weather conditions, anywhere on or near the Earth where there is an unobstructed line of sight to four or more GPS satellites.

GPS systems are extremely versatile and can be found in almost any industry sector. They can be used to map forests, help farmers harvest their fields, and navigate airplanes on the ground or in the air. GPS systems are used in military applications and by emergency crews to locate people in need of assistance. Hence with a wide variety of its usage the GPS technology to solve the problems faced by the fishermen. We will be using the concept of radar and find the area where the density of fishes is more and where it is less so that it won't be time consuming.

CHAPTER – 2

PROPOSED SYSTEM

2.1 HARDWARE AND COMPONENTS DESCRIPTION

A) Arduino Uno

The Arduino Uno is a microcontroller board which is developed by a company that goes by the name Arduino. It is an open-source microcontroller board and is based on ATmega328P microcontroller. The Arduino Uno has a set of digital as well as analog input and output pins that can be used in various circuits and expansion boards. It consists of 14 digital pins and 6 analog pins. It is programmable with Arduino IDE (Integrated Development Environment) using a type B USB cable. It works fine with voltages between 7volts and 20 volts, although it is mostly used with a 9-V battery. The name Uno is because it was among the first USB-based Arduino boards.

B) GNSS-

The Global Navigation Satellite System Global Navigation Satellite System (GNSS) is a huge collection of Satellite that provide signals from space and transmits position and timing related data to GNSS receivers. The receiver afterwards uses this data to determine locations. It can provide positioning to worldwide users on a continuous basis in all weather, irrespective of day or night, anywhere on or near the Earth provided the place has an unobstructed view of four or more GNSS satellites. GNSS satellites broadcasts signals from space that GNSS receivers uses to provide latitude, longitude and altitude. GNSS is made up of three segments: Space, Control and User. The Space Segment consists of 24 to 32 satellites in medium Earth orbit and also contains boosters required to launch them into orbit. The Control system is composed of a Master Control Station, an Alternate Master Control Station, and a host of dedicated and shared Ground Antennas and Monitor Stations. The User segment is composed of hundreds of thousands of allied military users of the secure GNSS Precise Positioning Service and tens of millions of civil, commercial and scientific users of the Standard Positioning Service. GNSS is widely being used worldwide for navigation purposes, tracking and surveillance.

C) Power Supply-

DC motor and Microcontroller requires some power source. This power is supplied using a DC power supply ranging from +5 V to +12 V. The available energy source is of 230v-50Hz, and a centre tapped step down transformer is used. The Arduino operates at low power.

D) Relay-

A Relay is an electrically operated switch. It consists of a set of input terminals for a single or multiple control signals, and a set of operating control terminals. Relays are used where it is needed to control a circuit by an independent low power signal. It is also used where it is needed to control several circuits by one signal. The conventional form of a relay uses an electromagnet to close or open the contacts. There are several other operating principles that have been invented, one of those are in solid state relays. It uses semiconductor properties for control and does not rely on moving parts.

E) Buzzer-

A Buzzer is an electronic device for signalling purposes. These are mostly used in automobiles, household appliances and other electronic devices. It has a number of switches connected to the control unit that determines if and which button was pushed, or if a present time has lapsed, and sounds a warning in the form of a continuous buzzing or beeping sound. If the boat approaches the restricted area, the buzzer will sound automatically and will keep on beeping by means of pulse width modulation.

F) LED-

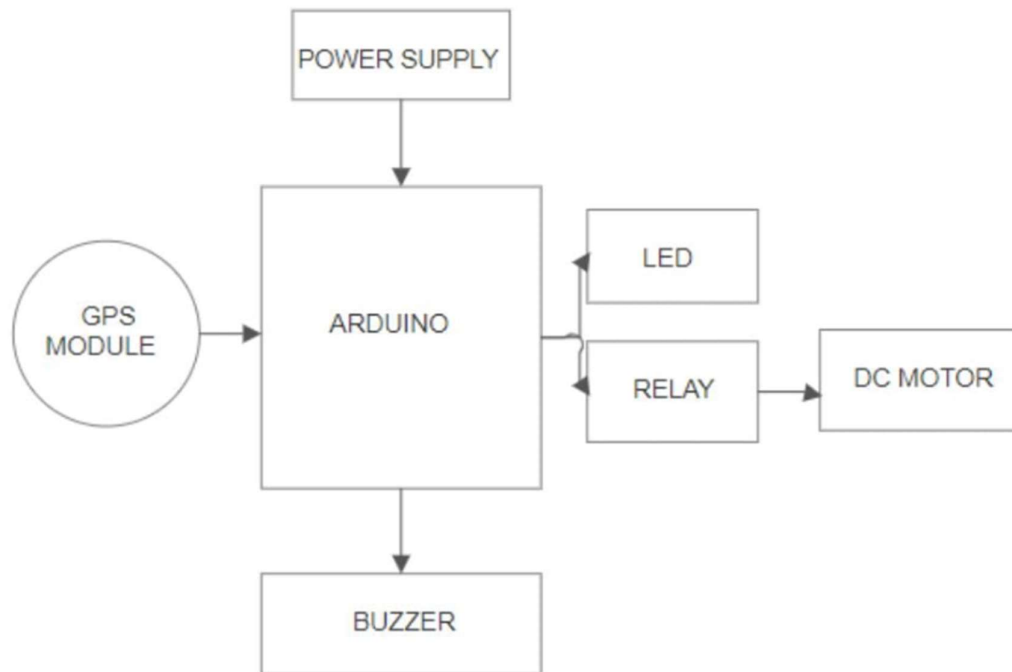
Connecting LED is just for an indication of whether it is near to the border or not.

Connect the positive terminal of the LED to the digital pin 11(D11).

Connect the negative terminal of the LED to the GND pin.

G) Ultra sonic sensor-

Waterproof ultrasonic sensing system for locating fish utilizes underwater detection technology with using acoustic devices (acoustic instruments) and operates with using SONAR (Sound Navigation, and Ranging). This technology use voice or sound for detection. It is known that the speed of sound in underwater is 1500 m / sec, while the speed of sound in free space is only 340 m / sec so the technology is very effective for detection under water



Block diagram for border alert system

2.2 Complete Setup of working module:

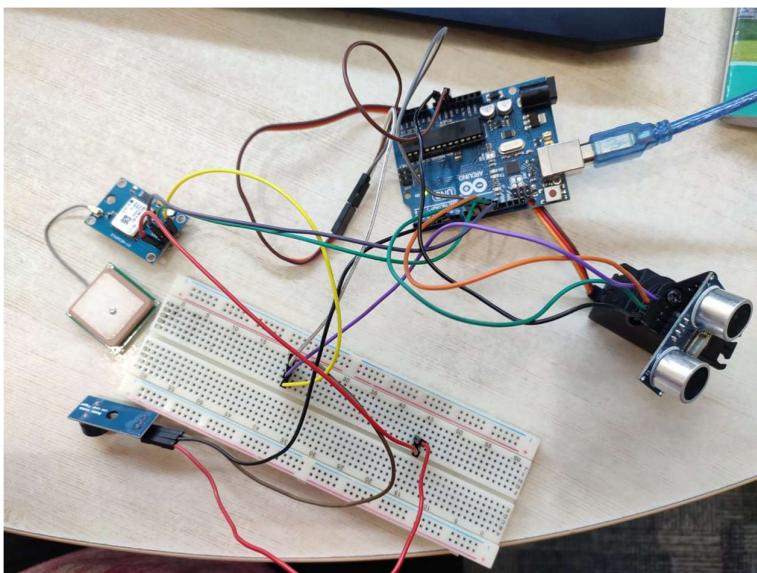


fig: the circuit of the proposed system

CHAPTER – 3

MODULE DESCRIPTION

Increasing tension across the borders cause many conflicts among countries and India, Sri Lanka is no exception. The sea border between countries is not easily identifiable. The problem of Indian fishermen unintentionally crossing the border and getting abducted remains unsolved. Further, the inevitable harsh climate and the path misleading has cost many lives. Based on a census, so far in 2017, 215 Indian fishermen have been apprehended. About 30% of the fishermen family has lost their beloved ones. This instructible tends to describe the role of various surveillance technologies such as radar, unmanned aerial vehicles, satellites that can play to avoid chaos.

We can design an embedded system that notifies the fishermen about borders using GPS (global positioning system). We use the GPS receiver to find the location of fishermen and the longitude, latitude values are sent to micro-controller. The fishermen are alerted a few meters before the borders. On further moving the engine will stop.

In this instructible, I will be showing you how to do the fisherman border alert system using Arduino. This was just a short introduction, everything else from theory, building, and programming will be explained in later steps.

Any instant of time, there are at least 4 GPS satellites in line of sight to a receiver on the earth. Each of these GPS satellites sends information about its position and the current time to the GPS receiver at fixed regular instants of time. This information is transmitted to the receiver in the form of a signal which is then intercepted by the receiver devices. These signals are radio signals that travel with the speed of light. The distance between a GPS receiver and the satellite is calculated by finding the difference between the time the signal was sent from the GPS satellite and the time the GPS receiver received the signal.

Once the receiver receives the signal from at least three satellites, the receiver then points its location using the trilateration process. A GPS requires at least 3 satellites to calculate 2-D position (latitude and longitude on a map). In this case, the GPS receiver assumes that it is located at the mean sea level. However, it requires at least 4 satellites to find receivers 3-D position (latitude, longitude, and altitude).

Arduino's processor basically uses the Harvard architecture where the program code and program data have separate memory. It consists of two memories- Program memory and data memory. The code is stored in the flash program memory, whereas the data is stored in the data memory. The Atmega328 has 32 KB of flash memory for storing code (of which 0.5 KB is used for the bootloader), 2 KB of SRAM and 1 KB of EEPROM and operates with a clock speed of 16MHz.

The Arduino Uno power supply can be done with the help of a USB cable or an external power supply. The external power supplies mainly include AC to DC adapter otherwise a battery. The adapter can be connected to the Arduino Uno by plugging into the power jack of the Arduino board. Similarly, the battery leads can be connected to the Vin pin and the GND pin of the POWER connector. The suggested voltage range will be 7 volts to 12 volts.

The program is done in Embedded C using Arduino software. the software link is attached below:

Arduino IDE Software

It requires two library files;

SoftwareSerial.zip

TinyGPS.zip

Install Arduino software and include the above Zip files and start coding. the code contains the fixed the longitude, latitude to which the current location is compared. Arduino code has two parts:

Void Setup (): where you can initialize the Pin modes

Void Loop (): The code inside this loop runs indefinitely.

During explanation the variation maybe within a few meters so the last decimal itself to be considered so approximate the value given by the GPS to show variance in meters. and the values returned by the GPS are latitude, longitude, altitude, direction, no of satellites connected but here we need only longitude and latitude. they are in float datatype.

3.2 CODE:

For border alert system:

```
#include<SoftwareSerial.h>
```

```
#include<TinyGPS.h>
```

```
SoftwareSerial SerialGPS(0,2);
```

```
TinyGPS GPS;
```

```
int led=11;
```

```
int flag=0;
```

```
float lat,lon,val;
```

```
float flat=11.02104*100000;
```

```
float flon=77.00417*100000;
```

```

const int buzzer = 8;

int i;

void setup()
{
    SerialGPS.begin(9600);

    pinMode(led,OUTPUT);

    pinMode(buzzer, OUTPUT);

    Serial.begin(9600);
}

void loop()
{
    while(SerialGPS.available())
    {
        if(GPS.encode(SerialGPS.read()))
        {
            GPS.f_get_position(&lat,&lon);

            Serial.println("-----");

            Serial.println("LINK TO GOOGLE MAPS:");

            Serial.print("http://maps.google.com/maps?q=");

            Serial.print(lat,5);

            Serial.print(",");

            Serial.print(lon,5);

```

```

Serial.println("");

lat=lat*100000;

lon=lon*100000;

Serial.print(lat,0);

Serial.print(",");

Serial.print(lon,0);

Serial.println("");

Serial.print(flat,0);

Serial.print(",");

Serial.print(flon,0);


if(lat>=flat)

{

//for(i=0;i<50;i++)

//{

digitalWrite(led,HIGH);

tone(buzzer, 1000); // Send 1KHz sound signal...

delay(100);      // ...for 1 sec

noTone(buzzer);  // Stop sound...

delay(100);      // ...for 1 sec

//}

if(lat>=flat+2)

{

// tone(buzzer, 1000);

```

```
}  
  
}  
  
else  
  
{  
  
    digitalWrite(led,LOW);  
  
}  
  
}  
  
}  
  
}
```

For fish detection:

```
#include<Servo.h>  
  
// set ouptut pins  
  
const int TriggerPin = 11;  
  
const int EchoPin = 12;  
  
const int motorSignalPin = 9;  
  
  
  
// starting loaction  
  
const int startAngle = 90;  
  
  
  
// rotation limits  
  
const int minimumAngle = 6;  
  
const int maximumAngle = 175;
```

```

// speed

const int degreesPerCycle = 1;


// Library class instance
Servo motor;


void setup(void)
{
    pinMode(TriggerPin, OUTPUT);

    pinMode(EchoPin, INPUT);

    motor.attach(motorSignalPin);

    Serial.begin(9600);
}


void loop(void)
{
    static int currentAngle = startAngle;

    static int motorRotateAmount = degreesPerCycle;


    // move motor

    motor.write(currentAngle);

    delay(10);

    // calculate the distance from the sensor, and write the value with location to serial

    SerialOutput(currentAngle, CalculateDistance());
}

```



```

// update motor location

currentAngle += motorRotateAmount;


// if the motor has reached the limits, change direction
if(currentAngle <= minimumAngle || currentAngle >= maximumAngle)
{
    motorRotateAmount = -motorRotateAmount;
}
}

```

```

int CalculateDistance(void)
{
    // trigger the ultrasonic sensor and record the time taken reflect
    digitalWrite(TriigerPin, HIGH);

    delayMicroseconds(10);

    digitalWrite(TriigerPin, LOW);

    long duration = pulseIn(EchoPin, HIGH);

    // convert this duration to a distance
    float distance = duration * 0.017F;

    return int(distance);
}

```

```

void SerialOutput(const int angle, const int distance)

```

```
{  
  
    // convert the angle and distance to a string and serial print  
  
    Serial.println(String(angle) + "," + String(distance));  
  
}
```

FOR FISH DETECTION IN PROCESSING:

```
import processing.serial.*;
```

```
import java.awt.event.KeyEvent;
```

```
import java.io.IOException;
```

```
Serial myPort;
```

```
PFont orcFont;
```

```
int iAngle;
```

```
int iDistance;
```

```
void setup() {
```

```
    size(1300, 800);
```

```
    smooth();
```

```
    myPort = new Serial(this, "COM6", 9600);
```

```
    myPort.clear();
```

```
    myPort.bufferUntil('\n');
```

```
    orcFont = loadFont("OCRAExtended-30.vlw");
```

```
}
```

```
void draw()
```

```
{
```

```
fill(98, 245, 31);

textFont(orFont);

noStroke();

fill(0, 4);

rect(0, 0, width, 0.935 * height);

fill(98, 245, 31);
```

```
DrawRadar();

DrawLine();

DrawObject();

DrawText();
```

```
}
```

```
void serialEvent (Serial myPort)
```

```
{
```

```
  try {
```

```
    String data = myPort.readStringUntil('\n');
```

```
    if (data == null) {
```

```
      return;
```

```
    }
```

```
    int commaIndex = data.indexOf(",");
```

```
    String angle = data.substring(0, commaIndex);
```

```
    String distance = data.substring(commaIndex+1, data.length()-1);
```

```
    iAngle = StringToInt(angle);
```

```
    iDistance = StringToInt(distance);
```

```

    } catch(RuntimeException e) {}
}

void DrawRadar()
{
    pushMatrix();

    translate(width/2, 0.926 * height);

    noFill();

    strokeWeight(2);

    stroke(98, 245, 31);

    // draws the arc lines

    DrawRadarArcLine(0.9375);

    DrawRadarArcLine(0.7300);

    DrawRadarArcLine(0.5210);

    DrawRadarArcLine(0.3130);

    // draws the angle lines

    final int halfWidth = width/2;

    line(-halfWidth, 0, halfWidth, 0);

    for(int angle = 30; angle <= 150; angle+=30) {

        DrawRadarAngledLine(angle);

    }

    line(-halfWidth * cos(radians(30)), 0, halfWidth, 0);

    popMatrix();

```

```

}

void DrawRadarArcLine(final float coefficient)

{
    arc(0, 0, coefficient * width, coefficient * width, PI, TWO_PI);
}

void DrawRadarAngledLine(final int angle){

    line(0, 0, (-width/2) * cos(radians(angle)), (-width/2) * sin(radians(angle)));
}

void DrawObject()

{
    pushMatrix();

    translate(width/2, 0.926 * height);

    strokeWeight(9);

    stroke(255,10,10);

    int pixsDistance = int(iDistance * 0.020835 * height);

    if(iDistance < 40 && iDistance != 0) {

        float cos = cos(radians(iAngle));

        float sin = sin(radians(iAngle));

        int x1 = +int(pixsDistance * cos);

        int y1 = -int(pixsDistance * sin);

        int x2 = +int(0.495 * width * cos);

        int y2 = -int(0.495 * width * sin);

        line(x1, y1, x2, y2);
    }
}

```

```

    }

    popMatrix();
}

void DrawLine()
{
    pushMatrix();

    strokeWeight(9);

    stroke(30, 250, 60);

    translate(width/2, 0.926 * height);

    float angle = radians(iAngle);

    int x = int(+0.88 * height * cos(angle));

    int y = int(-0.88 * height * sin(angle));

    line(0, 0, x, y);

    popMatrix();
}

void DrawText()
{
    pushMatrix();

    fill(0, 0, 0);

    noStroke();

    rect(0, 0.9352 * height, width, height);

    fill(98, 245, 31);

    textSize(25);

```

```

text("10cm", 0.6146 * width, 0.9167 * height);

text("20cm", 0.7190 * width, 0.9167 * height);

text("30cm", 0.8230 * width, 0.9167 * height);

text("40cm", 0.9271 * width, 0.9167 * height);


textSize(40);

text("Object: " + (iDistance > 40 ? "Out of Range" : "In Range"), 0.125 * width, 0.9723 * height);

text("Angle: " + iAngle + " °", 0.52 * width, 0.9723 * height);

text("Distance: ", 0.74 * width, 0.9723 * height);

if(iDistance < 40) {

    text("    " + iDistance + " cm", 0.775 * width, 0.9723 * height);

}

textSize(25);

fill(98, 245, 60);

translate(0.5006 * width + width/2 * cos(radians(30)), 0.9093 * height - width/2 *
sin(radians(30)));

rotate(-radians(-60));

text("30°",0,0);


resetMatrix();


translate(0.497 * width + width/2 * cos(radians(60)), 0.9112 * height - width/2 *
sin(radians(60)));

rotate(-radians(-30));

text("60°",0,0);

```

```
resetMatrix();
```

```
    translate(0.493 * width + width/2 * cos(radians(90)), 0.9167 * height - width/2 *  
sin(radians(90)));
```

```
    rotate(radians(0));
```

```
    text("90°",0,0);
```

```
    resetMatrix();
```

```
    translate(0.487 * width + width/2 * cos(radians(120)), 0.92871 * height - width/2 *  
sin(radians(120)));
```

```
    rotate(radians(-30));
```

```
    text("120°",0,0);
```

```
    resetMatrix();
```

```
    translate(0.4896 * width + width/2 * cos(radians(150)), 0.9426 * height - width/2 *  
sin(radians(150)));
```

```
    rotate(radians(-60));
```

```
    text("150°",0,0);
```

```
    popMatrix();
```

```
}
```

```
int StringToInt(String string)
```

```
{
```

```
    int value = 0;
```

```
    for(int i = 0; i < string.length(); ++i) {
```

```
        if(string.charAt(i) >= '0' && string.charAt(i) <= '9') {
```

```
            value *= 10;
```



```
    value += (string.charAt(i) - '0');  
  
    }  
    return value;}  

```

CHAPTER – 4

RESULTS AND DISCUSSION

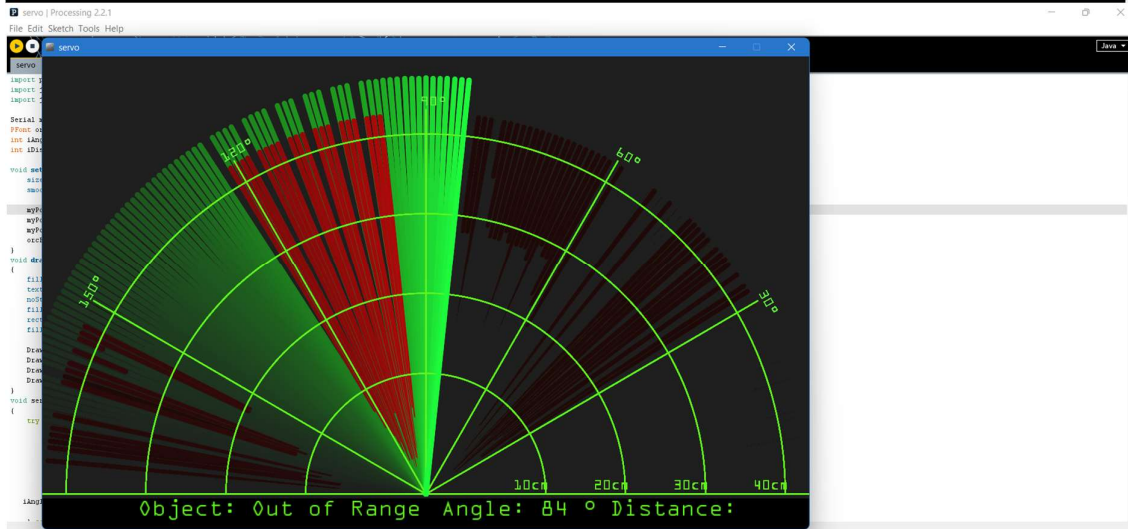
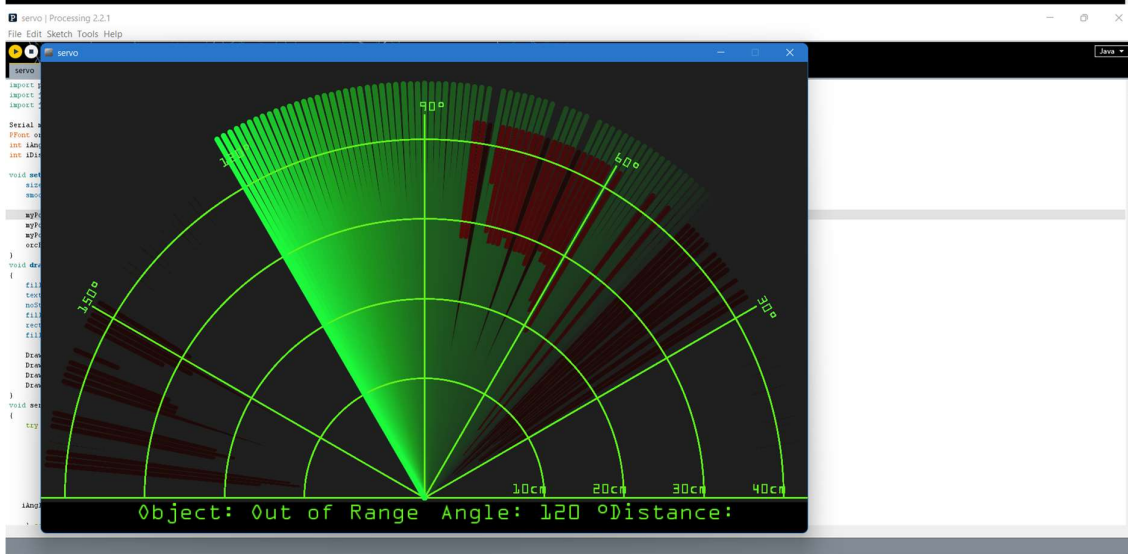
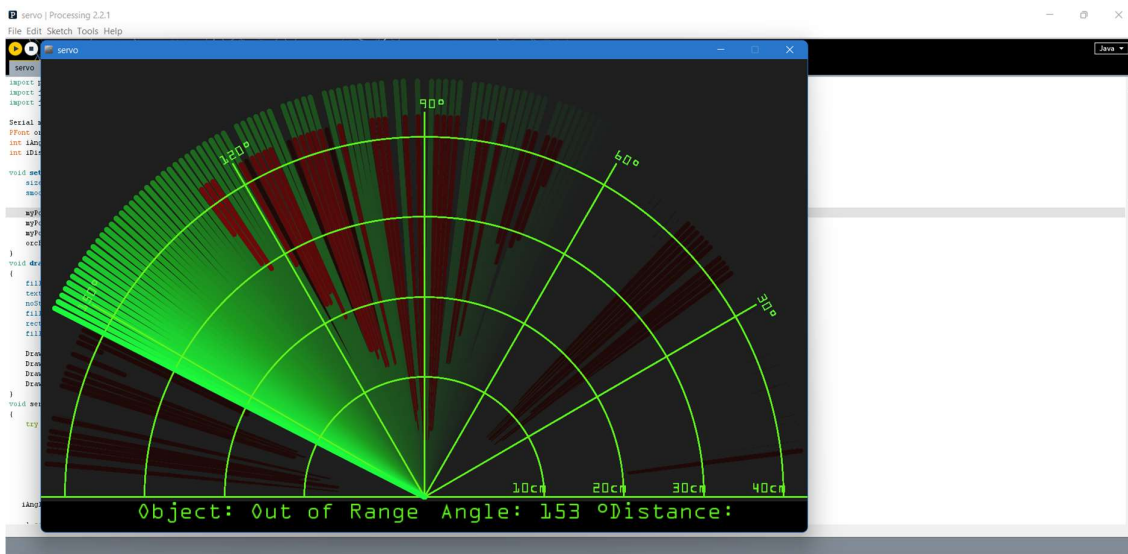
The system was successfully configured and tested with the specified coordinates, yielding positive results.

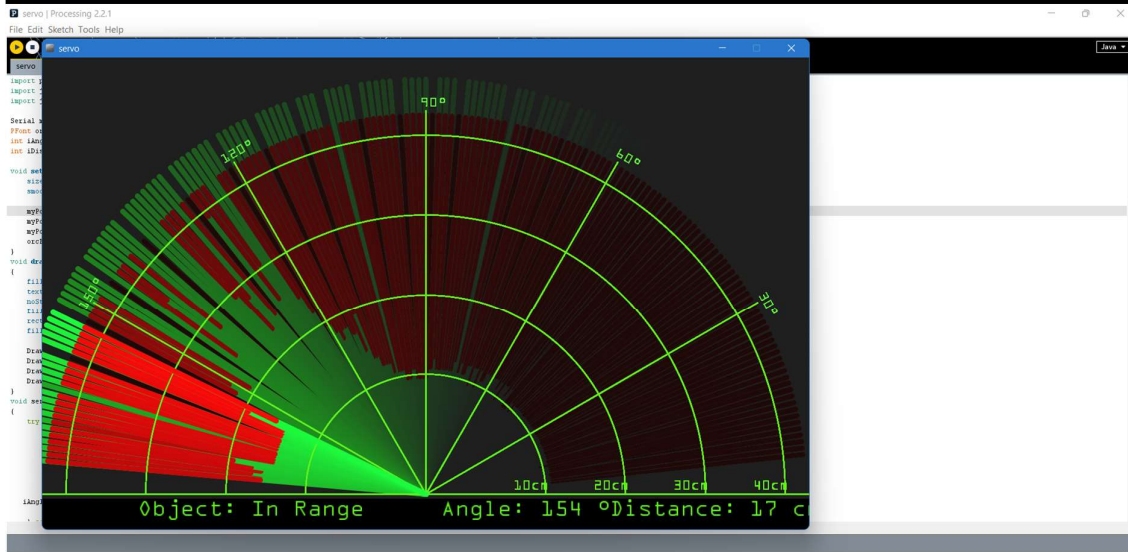
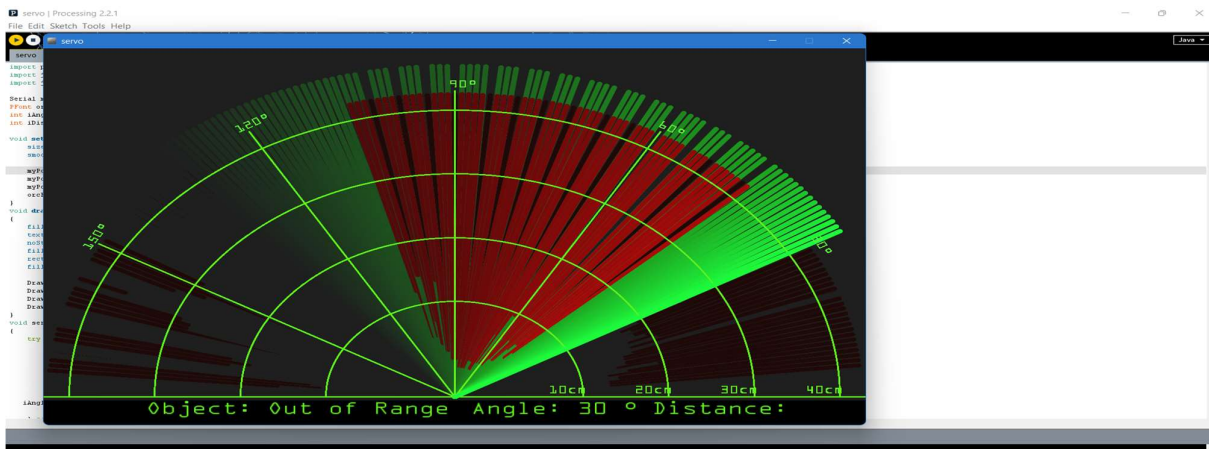
The position, which is the longitude and latitude, is depicted in the diagram below, along with a link to Google Maps.

If a fisherman crosses the border, a buzzer will sound to inform him.

```
COM6
|
|
LINK TO GOOGLE MAPS:
http://maps.google.com/maps?q=12.84179,80.15432
1284179,8015432
-----
LINK TO GOOGLE MAPS:
http://maps.google.com/maps?q=12.84179,80.15432
1284179,8015432
-----
LINK TO GOOGLE MAPS:
http://maps.google.com/maps?q=12.84179,80.15432
1284179,8015432
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LINK TO GOOGLE MAPS:
http://maps.google.com/maps?q=12.84179,80.15432
1284179,8015432
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LINK TO GOOGLE MAPS:
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1284179,8015433
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1284179,8015433
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LINK TO GOOGLE MAPS:
http://maps.google.com/maps?q=12.84179,80.15433
1284179,8015433
-----
LINK TO GOOGLE MAPS:
http://maps.google.com/maps?q=12.84179,80.15432
1284180,8015432
1284180,8015432
1284180,8015432
```

The figures below indicate the farmer how close an object is approaching him and how close it is, allowing him to keep alert if a dangerous fish is approaching.





CHAPTER – 5

CONCLUSION

Border detection system for fishermen using an Arduino and GPS module is the simplest way to find the border. The information comes from the GPS in the form of coordinates i.e., in the form of latitudes and longitudes. As many fishermen unknowingly cross the maritime border between India and Sri Lanka, this Navigation System will help them know their position and will allow them to avoid the consequences of crossing the border. The project is low budget and it will help the poor fishermen to a great extent. The fish detection using underwater ultrasonic sensors is the basic and simplest method of finding the fishes. The project helps the fishermen to find the fish in a quick and easy way without taking any risk of crossing the border to find the fish.

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